Conversion of Borneol to Camphor

Sarah Rogers

November 22, 2023

Chem 2081 007

Introduction

In this experiment, camphene was converted to camphor. A water bath was used to allow the reaction to take place. The collected product was placed on a watch glass and weighed at the end of the experiment.

Materials and Methods

On an analytical balance, 0.324 g of borneol was weighed in a long-neck 10 mL round-bottom flask. After that, 0.3 mL of glacial acetic acid was added to the round-bottom flask. Then, 1 mL of acetone and a boiling chip was added to the round-bottom flask containing the mixture. Next, an open reflux setup was assembled according to Figure 1-18 on page 20 of the CHEM 2081 Organic Chemistry II Lab Manuel (Keller, 2023). The apparatus was wrapped with a cold paper towel. The round-bottom flask was placed in a water bath with a temperature under 50°C. Once the reagents dissolved, 0.5 mL of sodium hypochloride solution was added dropwise each minute for about 12 minutes. Each time hypochloride was added, the mixture was given a swirl. The reaction proceeded for about 15 minutes. Once the mixture cooled, the apparatus was disassembled. Next, the mixture was poured into a centrifuge tube. After that, 2 mL of methylene chloride was added to the centrifuge tube using a pipette. Then, about 5 mL of sodium bicarbonate was added to the centrifuge tube and the bottom layer was washed over the top layer several times. Next, the organic layer was transferred to a beaker and the top layer was transferred to a flask. After that, the organic layer was transferred back into the centrifuge tube and washed with 2 mL of sodium bisulfite with 2 mL of water. The organic layer was then transferred to a beaker and the top layer was transferred to a flask. Then, the organic layer was

transferred back into the centrifuge tube and a spatula tip's worth of anhydrous sodium sulfate was added. After that, an empty watch glass was weighed on an analytical balance. Then, the solution was transferred to the dry watch glass and placed on a water bath for several minutes. After that, the pre-weighed watched glass was placed on an analytical balance and the weight was recorded. Finally, the glassware was cleaned with soap and water inside the hood.

Results

Weight of Dry Watch Glass (g)	37.128
Weight of Watch Glass with Crystals (g)	37.323
Weight of Crystals (g)	0.195

Calculations

Moles of Borneol: $0.324\ g\ \times \frac{1\ mol}{154.25\ g} =\ 0.00210049\ mol$

Grams of Camphor: $0.00210049\ mol \times \frac{152.23\ g}{1\ mol} =\ 0.31975702\ g$

Percent Yield: $\frac{0.195\ g}{0.320\ g} \times 100 =\ 60.9\%$

Discussion

In this experiment, borneol was converted to camphor. At the end of the experiment, the mass of camphor was determined by weighing the product on a watch glass. It is known that the product on the watch glass was borneol because of the formation of separate layers. In this experiment, the bottom layer contained the organic product. It is known that the organic product was in the bottom layer because of density. Water has a density of 1 g/mL, and methylene chloride has a density of 1.33 g/mL, so the camphor would be in the bottom layer. At the end of the experiment, 0.195 g of product was recovered. The percent yield of camphor was found to be 60.9%. This means that some of the camphor was not recovered. One error that could have

occurred is transferring the drying agent, anhydrous sodium sulfate, onto the watch glass. When transferring the product onto the watch glass, only the camphor should have been transferred from the centrifuge tube. Transferring the drying agent would have caused the percent yield to be overestimated because there would be additional product on the watch glass. Another error that could have occurred involved losing solvent through evaporation. If the distillation column is not kept cool enough during reflux, some product could be lost. To help prevent this error from occurring in this experiment, a cold paper towel was wrapped around the apparatus. Even though a cold paper towel was wrapped around the apparatus, this was most likely the error that occurred in this experiment. This is because the temperature of the water bath kept exceeding 50°C during reflux. This means that the temperature of the mixture in the round-bottom flask was most likely too high. To help prevent this error from occurring, the rheostat should be set to a lower setting to keep the sand bath from reaching too high of a temperature. In this experiment, it was difficult to control the temperature of the water bath since the sand bath was set to a higher setting at the beginning of the experiment.

References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry II Lab Manual [lab manual]. Auburn (Al): Auburn University.